Application of Biso Enzyme in swine production

Biso enzyme is mainly utilized in pig farms across Japan for purposes such as deodorization, pest management, and composting. This document will outline the anticipated effectiveness and performance for each application, highlight key points, provide usage instructions, dosage, and other relevant information, so we appreciate your patience.

1. Odor elimination and pest management

The smell emanating from pig farms primarily comprises ammonia (NH4), low fatty acids (R-COOH), sulfur compounds (H2S), and various other odorous elements, including the putrid scent associated with manure. Notably, the ammonia scent often intensifies during the processing of manure.

Biso enzymes decompose or modify these odor-producing components through a chain-catalytic action involving multiple enzymes, thereby providing a deodorizing effect. Specifically, the previously mentioned odor-producing components are associated with H (hydrogen), and enzymes that primarily facilitate hydrogen separation reactions target H to separate or transfer it, thus breaking down or changing the odor. This process dim~~inishes the unpleasant smell of ammonia and other substances.~~

T~~he primary methods employed are "a. space spraying" and "b. cleaning~~".

When aerosolized, the typical dilution ratio is between 1,000 and 2,000 times. In certain instances, it may be diluted as much as 10,000 times.

In an ideal scenario, one could consistently aerosolize diluted products into the atmosphere; however, this approach necessitates a significant volume of water and is not financially viable.

Consequently, deodorization is accomplished by consistently spraying the air for a specific duration.

For instance, if utilized to eliminate odors within a pigsty, the scent, similar to ammonia generated from waste, would permeate the space, and the concentration would steadily rise.

An increase in concentrations can result in diminished pig productivity and quality, stemming from compromised pig health, heightened stress levels, a deteriorating management and working environment, and adverse effects on worker health.

Consequently, it is essential to consistently prevent indoor concentrations from increasing, and routinely misting diluted Biso Enzyme into the air serves as an effective approach to accomplish this.

It also aids in managing pests and diseases.

Please consult the following example for usage amounts.

(Reference example)

Pig barn area: 20 × 100m = 2,000m²/building. Spray capacity and number of nozzles: 0.05 ℓ/minute x 100/nozzle = 5 ℓ/minute. Spraying time and interval: Functions for 1 minute per cycle, followed by a 5-minute pause between each spray.

Number of occurrences per hour: 60 minutes ÷ (1 minute + 5 minutes) = 10 occurrences/hour. Total water volume over 12 hours of operation:
10 times/hour x 12 hours x 5 liters/minute = 600 liters/day

If the dilution ratio of Biso Enzyme is 2,000, then 600 liters per day ÷ 2,000 = 0.3 liters/day

Selecting equipment, including spray nozzles and electric pumps, is essential. Additionally, it is important to consider the spraying time and dilution ratio, taking into account site conditions, odor intensity, and budget.

▮Biso enzyme and overall water consumption under the specified conditions

Pigpen area	Biso Enzyme (2,000 times diluted)	Totalwater consumption
1,000m2	0.15 ℓ/day	300 liters/day
2,000m2	0.3 liters/day	600 ml/day
5,000m2	0.75 ℓ/day	1,500 liters/day
10,000m2	1.5 liters/day	3,000 liters/day
20,000m2	3.0 liters/day	6,000 ℓ/day

b. Cleaning involves washing the floors, walls, equipment, and the bodies of the pigs within the pigsty.

In locations where it is challenging to spray into the air, it is advisable to utilize a diluted Biso Enzyme solution as a cleaning agent. The intensity of odors within the pig pen can be diminished by routinely cleaning the floors, walls, equipment, and the pigs' bodies on a daily basis or at regular intervals.

The typical dilution ratio for cleaning is between 5,000 and 10,000 times. Nevertheless, in certain situations, it can be diluted as much as 20,000 times.
Please consult the following example for usage amounts.

(Reference example)

Pig pen area: 20 × 100m = 2,000 m²/building. Water usage for cleaning: 1 liter/m² x 2,000 m² = 2,000 liters/building.

If the dilution ratio of Biso Enzyme is 10,000 times, then 2,000 liters divided by 10,000 equals 0.2 liters for one building.

Furthermore, cleaning the especially soiled sections of the pig pen is also effective. In this scenario, additional water can be conserved.

2. Encouraging the Composting Process: The composting process is conducted using manure gathered from pig farms.

The raw material, manure, is rich in organic matter, serving as a source of carbon and nitrogen. The composting process initiates when the moisture content is elevated.

During the primary fermentation phase, specifically from the sugar breakdown stage to the cellulose breakdown stage, it is essential to actively enhance the quantity and activity of aerobic bacteria, thereby accelerating the decomposition of organic matter. To achieve this, consistently spray and dilute Biso Enzyme water.

The primary processes of composting and the roles of common bacteria are outlined below.

	Composting process:	Notes column	General role of fungi
1.	Introducing organic materials;	Enzyme mixture,	(Low temperature phase) Role of sugar-degrading bacteria
2.	Adjusting moisture levels.	Biso-supplementary-materials,and recycled organic fertilizer.	
3.	Initial fermentation	Mix, supply oxygen, carry out aerobic decomposition, and apply Biso enzyme solution (once a week, for a minimum of 5 applications).	(Medium to high temperature phase) Role of hemicellulose and cellulose-degrading bacteria (Medium temperature phase)
4.	Secondary fermentation	The process of	Role of lignin-degrading bacteria
5.	product finalization	aging is nearing complete maturity.	

When composting, the typical dilution ratio is between 1,000 and 2,000 times.

However, in certain instances, it can be diluted as much as 10,000 times.

When preparing the compost mixture, dilute the product between 1,000 and 2,000 times with water and apply it evenly using a fine mesh spray bottle. The suggested quantity of water is 2 to 4 liters for every 1 ton of raw organic fertilizer. In simpler terms, the product usage translates to 2 to 4 liters/m³ ÷ 1,000 to 2,000 times = 1 to 4 ml/m³(t).

Please consult the following example for the appropriate amount to utilize.

(Reference example)

Manure output per pig: 2.0 kg/day

Total number of animals: 2,000

Desired daily manure volume:

2.0 kg/day x 2,000 animals = 4,000 kg = 4 cubic meters/day

*If the necessary quantity of Biso Enzyme is 1 ml/m3, then the quantity required for the intended daily fertilizer application is 4m3/day x 1 ml/m3 = 4 ml/day (this represents the volume of undiluted product).

*Then, dilute 4 ml of the initial solution each day with water at a ratio of 1,000 to 2,000 times and spray it onto the raw fertilizer applied daily. 4 ml/day x 1,000 to 2,000 times = 4,000 to 8,000 ml = 4 to 8 liters/day

*Furthermore, during the primary fermentation stage of the composting process, organic fertilizer has been generated and is currently being produced. Keep track of the quantity of compost you have gathered and add more on a weekly basis.

▮ Quantity of Biso Enzyme utilized under the aforementioned conditions

Number of items	Enzyme Biso (1 mL/m³)	When utilized once a week, five times a week.
1m3(t)	1ml/m3	5ml
10m3 (t)	10ml/m3	50ml
100m3 (t)	100ml/m3	500ml
1,000 m³ (t)	1,000ml/m3	5,000ml

It is advisable to apply the mixture while rotating or mixing the manure pile.

When using, dilute with water at a ratio of 1,000 to 2,000 times and spray the resulting mixture.

Factors such as the quantity utilized, usage frequency, dilution ratio, and others must be taken into account according to site conditions, odor strength, budget constraints, humidity levels, and cutting and machining durations.

[Anticipated cost-efficiency]

*By decreasing the levels of odors and harmful gases,

Anticipated enhancements in pig productivity (enhanced survival rate, weight gain rate, feed conversion efficiency, etc.); decrease in flies and other disease-transmitting insects; better working conditions and diminished strain on employees; prevention of structural and equipment corrosion; initiatives for the surrounding environment and local residents.

Enhances the fermentation process of organic fertilizer while minimizing odors.

High-quality organic fertilizers contain a dense population of microorganisms and enzymes, leading to significant added value. They promote plant growth, decrease production time, enhance yield, and minimize odors during fermentation and transportation.

Monitor the decrease of toxic gas odors.

(This translation is derived from the content of the test report)

1. Testing Institute: Institute of Odor Experts in Korea

An approved trial was carried out at a research institute.
Consult research specialists in odor control to verify the efficacy of odor reduction.

3. Product utilized: Biso deodorizing enzyme (hereinafter referred to as the enzyme)

4. Testing methods: Detailed according to the accompanying test report.
(translated)

Dilute the liquid sample (enzyme) supplied by the customer 50 times with
Add 20 ml of distilled water to an empty 5 L distribution flask and seal it tightly.

2. Introduce the test gas at an initial concentration of 50 μmol/mol (PPM) and assess the concentration.

Initial test gas concentration (0 minutes) and after 30 minutes. This is known as the sample concentration (enzyme addition).

Three test gas concentrations were measured in accordance with KSI 2218:2009.

Maintain the temperature at 23°C ± 5°C and the humidity at 50% ± 10% during evaluation procedure.

5 Conduct tests akin to those in section 2 to 4 above without any example. This is referred to as the blank concentration.

o ⑥ Evaluate the gas removal rate for each time interval in the subsequent equation:

Test gas removal rate (%) = {(Bag concentration – Sample concentration)} x 100

*Note: 1 μmol/mol is equivalent to
1 PPM (mg/L)

5. Kết quả thử nghiệm

(1) Hiđrô sunfua(H2S)

項目		試験結果			試験方法
		Blank 濃度 (μmol／mol)	Sample 濃度 (μmol／mol)	除臭率 (%)	
硫化水素 H2S	0（時間）	50	50	0.0	委託
	0.5（時間）	49	36	26.5	
	1（時間）	49	32	34.7	
	3（時間）	48	27	43.8	
	5（時間）	47	21	55.3	
	12（時間）	45	10	77.8	
	15（時間）	45	6	86.7	
	18（時間）	45	3	93.3	
	24（時間）	44	0	100.0	

※検出限界　0.1 μmol／mol

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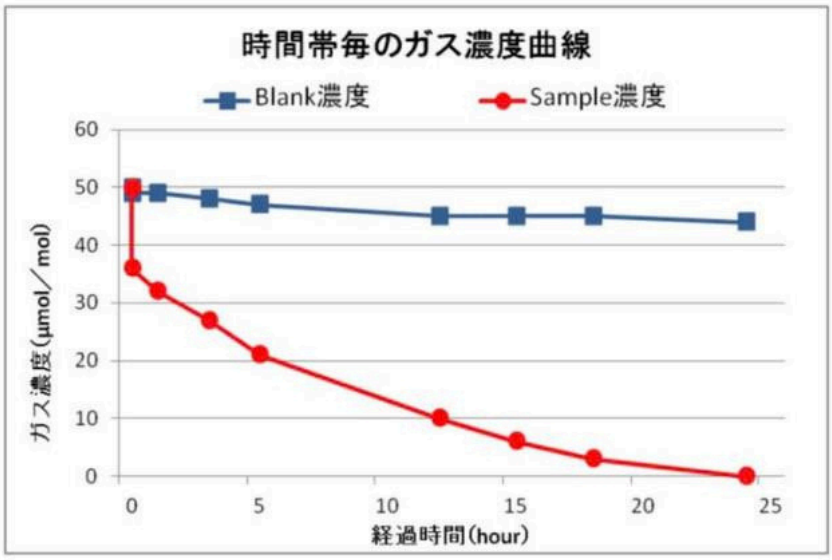

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株式会社　美創技研

※Báo cáo thử nghiệm (bản gốc) Hydro sunfua(H2S)

the way to trust

시험성적서 硫化水素

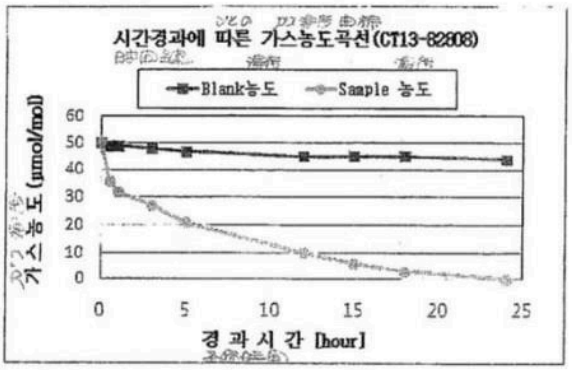
試驗成績書

성적서번호 : CT13-82808

6. 시험결과

시험항목		시험결과			시험방법
		Blank농도 (μmol/mol)	Sample농도 (μmol/mol)	탈취율 (%)	
탈취 시험 황화수소 H_2S	0 (시간)	50	50	0.0	의뢰자제시방법
	0.5 (시간)	49	36	26.5	
	1 (시간)	49	32	34.7	
	3 (시간)	48	27	43.8	
	5 (시간)	47	21	55.3	
	12 (시간)	45	10	77.8	
	15 (시간)	45	6	86.7	
	18 (시간)	45	3	93.3	
	24 (시간)	44	0	100.0	

※ 검출한계 0.1 μmol/mol



※ 시험방법

1. 의뢰자가 제시한 액상시료를 증류수로 50배 희석한 후 20 ㎖를 5 L 크기 반응기에 넣고 밀봉함.
2. 시험가스의 초기농도를 50 μmol/mol으로 주입하고 시험가스의 농도를 초기(0분), 30분 외 시간에서 측정하고 이를 sample 농도라 함.
3. 시험가스의 농도는 KS I 2218:2009에 의해 측정함.
4. 시험 중 온도는 23 ℃ ± 5 ℃, 습도는 50 % ± 10 %를 유지함.
5. 이와 별도로 시료가 없는 상태에서 위의 2 ~ 4에 의해 시험을 진행하고 이를 blank 농도라 함.
6. 각 시간대별 시험가스의 제거율은 다음 식에 의해 계산함.

시험가스의 제거율(%) = [{(blank 농도)-(sample 농도)}/(blank 농도)] × 100.끝.

—— 이 하 여 백 ——

양식QP-20-01-06(0)

（2）Amoniac(NH3)

項目		試験結果			試験方法
		Blank 濃度（μmol／mol）	Sample 濃度（μmol／mol）	除臭率（%）	
アンモニア NH3	0（分）	50	50	0.0	委託
	30（分）	49	0	100.0	
	60（分）	49	0	100.0	
	90（分）	49	0	100.0	
	120（分）	48	0	100.0	

※検出限界　0.2μmol／mol

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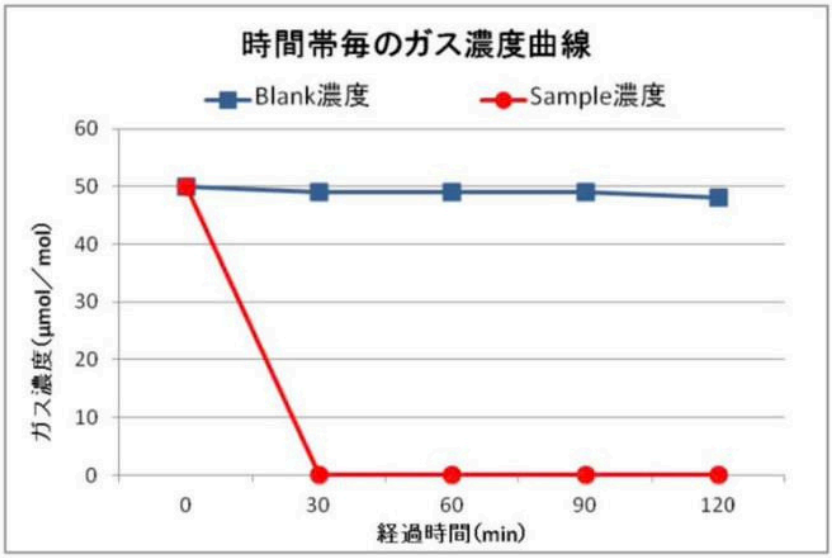


※Báo cáo thử nghiệm (bản gốc) Hydro sunfua(NH3)

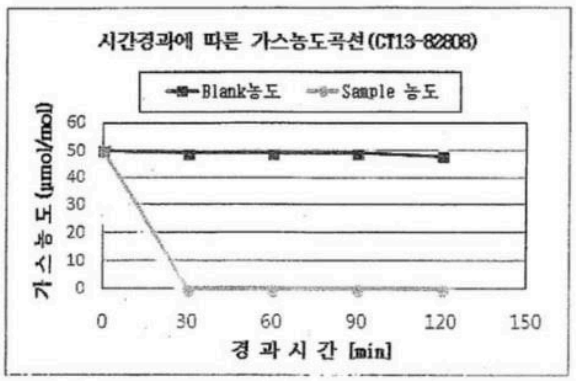
시험성적서

성적서번호 : CT13-82808

6. 시험결과

시험항목		시험결과			시험방법
		Blank농도 (μmol/mol)	Sample농도 (μmol/mol)	탈취율 (%)	
탈취 시험 암모니아 NH_3	0 분	50	50	0.0	의뢰자제시방법
	30 분	49	0	100.0	
	60 분	49	0	100.0	
	90 분	49	0	100.0	
	120 분	48	0	100.0	

※ 검출한계 0.2 μmol/mol



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